Exact current-bit optimum and a robust history advantage for coarse-grained generalized-amplitude-damping feedback

Xiaojian Zeng
PBC School of Finance, Tsinghua University, Beijing 100190, China
xiaojianzeng@foxmail.com

Abstract

Classical feedback from a measured environment is usually analysed with every Kraus outcome resolved. We study two uses of a full-Choi-rank generalized-amplitude-damping qubit instrument whose four outcomes are disclosed only through the bit 03|12. For $p = 18/25$ and $\gamma = 3/10$, arbitrary clock-dependent current-bit CPTP recoveries have exact maximal entanglement fidelity $289/400$, attained by identity recovery. A feasible controller retaining the first bit reaches $1831/2500$, an advantage of $99/10000$ over the current-bit optimum. We also certify a rational parameter rectangle on which the full-history advantage is uniformly at least $1/1000$.

Keywords: environment-assisted correction; quantum feedback; coarse-grained measurement; entanglement fidelity; generalized amplitude damping

1 Introduction

Measuring an environment converts a quantum noise process into a classical record and a quantum instrument. When each record value resolves one Kraus operator, conditional recovery can exploit its polar decomposition. Gregoratti and Werner developed this discrete-time framework and obtained exact qubit recovery formulas for resolved records [1]. More recently, Rexiti, Khabbazi Oskouei, and Mancini compared current-outcome and history-conditioned feedback in the same resolved-record setting [2]. Related Bayesian–Markovian distinctions also occur in continuously monitored feedback, although there the dynamics and objective are different [3].

A finite classical readout need not resolve the environment outcome. If several orthogonal outcomes are assigned the same symbol, the disclosed branch is a multi-Kraus completely positive operation: the hidden alternatives must be added incoherently before recovery. Consequently, the controller cannot apply a separate polar correction to each hidden Kraus operator. This information restriction is distinct from optimizing a single recovery after fixed noise, which is a semidefinite program [4], and from monotone alternating optimization of encoding and decoding maps [5]. The two-stage problem considered below is bilinear even though either stage becomes convex when the other is fixed.

We give an exact analysis for two uses of a standard generalized-amplitude-damping (GAD) qubit channel [6]. Its ordered four-outcome environment measurement is coarse grained into the disclosed bit 03|12. The controller knows the clock and the current bit but retains no earlier record. At the reference point $(p, \gamma) = (18/25, 3/10)$, we prove a sharp global upper bound over four arbitrary, possibly nonunital, qubit completely positive trace-preserving (CPTP) recovery maps. The bound is attained by identity recovery and gives

$$F_2^M = \frac{289}{400}. \tag{1}$$

An explicit controller that retains the first disclosed bit reaches 1831/2500. Thus its advantage over the optimal current-bit controller is exactly 99/10000, and a history advantage occurs at the earliest possible depth, two uses. We then prove a uniform, all-CPTP separation on the closed rectangle

$$\mathsf{R} = \left[\frac{71}{100}, \frac{723}{1000}\right] \times \left[\frac{29}{100}, \frac{303}{1000}\right]. \tag{2}$$

The rectangle is a sufficient certified region, not a maximality statement. We do not determine the exact full-history optimum, exact off-centre current-bit optima, or the behaviour of other environment pairings.

2 Instrument, policies, and fidelity

Let $0 < p < 1$, $q = 1 - p$, $0 < \gamma < 1$, and $s = \sqrt{1-\gamma}$. In the ordered energy basis $(|0\rangle, |1\rangle)$, fix the Kraus tuple

$$\begin{aligned} K_0 &= \sqrt{p}\ \mathrm{diag}(1, s), \quad K_1 = \sqrt{p\gamma}\,|0\rangle\langle 1|, \\ K_2 &= \sqrt{q}\ \mathrm{diag}(s, 1), \quad K_3 = \sqrt{q\gamma}\,|1\rangle\langle 0|. \end{aligned} \tag{3}$$

The ordering is part of the physical specification: the environment is measured in this fixed minimal Kraus basis. Direct summation gives $\sum_x K_x^\dagger K_x = I$. The two off-diagonal operators occupy distinct matrix units, while the squared determinant of the coordinate vectors of K_0, K_2 in the diagonal subspace is $pq\gamma^2$. Hence the four operators are linearly independent whenever $p, q, \gamma > 0$, and the channel has Choi rank four.

Only a balanced bit is disclosed. With

$$G_0 = \{0, 3\}, \qquad G_1 = \{1, 2\}, \tag{4}$$

the observed instrument is

$$\mathcal{I}_b(\rho) = \sum_{x \in G_b} K_x \rho K_x^\dagger. \tag{5}$$

The fine label x is summed incoherently before control. Two identical channel uses have fresh, independent environments.

For a current-bit policy, choose an arbitrary qubit CPTP map $\mathcal{R}_{k,b}$ for each clock $k \in \{1, 2\}$ and disclosed bit $b \in \{0, 1\}$. The clock is available, so $\mathcal{R}_{1,b}$ and $\mathcal{R}_{2,b}$ need not agree. At the second use the controller does not know the first bit, either hidden Kraus label, or any retained quantum state. The unconditional corrected channel factorizes exactly:

$$\Phi_{\boldsymbol{R}}^M = \mathcal{C}_2 \circ \mathcal{C}_1, \qquad \mathcal{C}_k = \sum_{b=0}^{1} \mathcal{R}_{k,b} \circ \mathcal{I}_b. \tag{6}$$

This is a time-inhomogeneous current-bit class, not a stationary-policy restriction. A full-history policy may instead use $\mathcal{R}_{2,b_1 b_2}$.

We use entanglement fidelity for the maximally mixed qubit,

$$F_e(\Phi) = \langle \Omega | (\mathrm{id} \otimes \Phi)(|\Omega\rangle\langle\Omega|) | \Omega \rangle, \qquad |\Omega\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}. \tag{7}$$

Let F_2^M and F_2^B denote the maxima over the current-bit and full-history classes. Compactness of the finite products of qubit-CPTP Choi sets ensures that both maxima are attained.

For later use, write a trace-preserving qubit map in affine Bloch form $\boldsymbol{r} \mapsto \boldsymbol{t} + T\boldsymbol{r}$. Its entanglement fidelity is

$$F_e(\Phi) = \frac{1 + \mathrm{Tr}\, T}{4}. \tag{8}$$

The affine translation does not enter the superoperator trace. Consequently, for the composition in Eq. (6),

$$F_e(\mathcal{C}_2 \circ \mathcal{C}_1) = \frac{1 + \mathrm{Tr}(T_2 T_1)}{4}. \tag{9}$$

3 A feasible retained-history controller

Let X be the Pauli bit flip. Consider the parameter-independent policy

$$R_{1,0} = I, \qquad R_{1,1} = X, \qquad R_{2,b_1 b_2} = \begin{cases} I, & b_1 = 0, \\ X, & b_1 = 1. \end{cases} \tag{10}$$

The second recovery retains b_1 but happens not to use b_2. All recoveries are unitary, so the policy is physically admissible. Expanding the incoherent branches only for evaluation gives

$$F_{\mathrm{hist}} = \frac{1}{4} \sum_{x_1, x_2 = 0}^{3} \left| \mathrm{Tr}\left[R_{2,g(x_1)g(x_2)} K_{x_2} R_{1,g(x_1)} K_{x_1} \right] \right|^2. \tag{11}$$

The nonzero trace amplitudes form the multiset

$$\begin{aligned} &p(2-\gamma), \quad 2\sqrt{pq(1-\gamma)}, \quad \gamma\sqrt{pq}, \quad p\gamma, \\ &\sqrt{pq}(2-\gamma), \quad 2q\sqrt{1-\gamma}. \end{aligned} \tag{12}$$

Squaring and simplifying yields the exact formula

$$H(p,\gamma) := F_{\mathrm{hist}} = 1 - \gamma + \frac{p\gamma^2}{2}. \tag{13}$$

At the reference point this is $H = 1831/2500$. Equation (13) is a feasible value, so $F_2^B \geq H$; it is not asserted to be the exact full-history optimum.

4 Exact current-bit optimum at the reference point

We now set

$$p = \frac{18}{25}, \qquad q = \frac{7}{25}, \qquad \gamma = \frac{3}{10}, \qquad r = 1 - \gamma = \frac{7}{10}, \qquad c = \frac{\gamma}{2} = \frac{3}{20}. \tag{14}$$

4.1 Branch contraction constraints

For $x \in \mathbb{R}^3$, direct evaluation of Eq. (5) on the traceless Hermitian input $x \cdot \boldsymbol{\sigma}$ gives scalar/Bloch pairs

$$\mathcal{I}_0(x \cdot \boldsymbol{\sigma}) : (c x_z, D_0 x), \qquad \mathcal{I}_1(x \cdot \boldsymbol{\sigma}) : (-c x_z, D_1 x), \tag{15}$$

where

$$\begin{aligned} d_0 = p\sqrt{r}, \qquad d_1 = q\sqrt{r}, \\ z_0 = p - c = \frac{57}{100}, \qquad z_1 = q - c = \frac{13}{100}, \qquad D_b = \mathrm{diag}(d_b, d_b, z_b). \end{aligned} \tag{16}$$

Here (a, v) denotes the operator $aI + v \cdot \boldsymbol{\sigma}$.

After arbitrary CPTP recoveries, let $B_b x$ be the Bloch vector of $\mathcal{R}_b \circ \mathcal{I}_b(x \cdot \boldsymbol{\sigma})$. This definition includes the recovery's nonunital translation multiplied by the branch scalar $\pm c x_z$. Trace preservation leaves the scalar component unchanged, and positivity implies trace-norm contraction on Hermitian inputs. Since

$$\|aI + v \cdot \boldsymbol{\sigma}\|_1 = 2 \max\{|a|, \|v\|_2\}, \tag{17}$$

we obtain

$$\|B_0 x\|_2 \le \|D_0 x\|_2, \tag{18}$$

$$\|B_1 x\|_2^2 \le \max\left\{d_1^2(x_x^2 + x_y^2) + z_1^2 x_z^2, c^2 x_z^2\right\}. \tag{19}$$

For Eq. (18), $\|D_0 x\|_2 \ge |c x_z|$ at the reference point. Thus $B_0^T B_0 \preceq D_0^2$, and, because D_0 is invertible,

$$B_0 = C D_0 \quad \text{for some real } C \text{ with } \|C\|_{\text{op}} \le 1. \tag{20}$$

4.2 A sharp Gram and nuclear-norm bound

Let $G = B_1^T B_1$. Average G over rotations about the input z axis. The right-hand side of Eq. (19) is rotation invariant, so the average remains feasible and has the form $\overline{G} = \operatorname{diag}(a, a, b)$. The averaging preserves $\operatorname{Tr} G$. It is also valid for the nuclear-norm estimate below: the function

$$G \longmapsto \operatorname{Tr} \sqrt{D_0 G D_0} \tag{21}$$

is concave on positive semidefinite matrices, while D_0 commutes with the rotations. Therefore its value at $\overline{G}$ is at least its value at G.

For a unit vector, put $t = x_z^2$. The two linear pieces on the right-hand side of Eq. (19) cross at

$$t_0 = \frac{d_1^2}{d_1^2 + c^2 - z_1^2} = \frac{49}{54}. \tag{22}$$

Checking the endpoints and the crossing on their respective linear intervals gives precisely

$$0 \le a \le d_1^2, \qquad 0 \le b \le c^2, \qquad 5a + 49b \le 49c^2. \tag{23}$$

The reference parameters obey the exact identity

$$5d_1^2 + 49z_1^2 = 49c^2. \tag{24}$$

Because $2 - 5/49 > 0$, Eqs. (23) and (24) imply

$$\|B_1\|_F^2 = \operatorname{Tr} G \le 2d_1^2 + z_1^2. \tag{25}$$

The same symmetrization and concavity give

$$\|B_1 D_0\|_* = \operatorname{Tr} \sqrt{D_0 G D_0} \le 2d_0\sqrt{a} + z_0\sqrt{b}. \tag{26}$$

The right side increases with b, so Eq. (23) reduces the remaining maximization to

$$f(a) = 2d_0\sqrt{a} + z_0\sqrt{c^2 - \frac{5a}{49}}, \qquad 0 \le a \le d_1^2. \tag{27}$$

Its derivative decreases on the open interval, and at the right endpoint

$$f'(d_1^2) = \frac{d_0}{d_1} - \frac{5z_0}{98z_1} = \frac{18}{7} - \frac{285}{1274} = \frac{2991}{1274} > 0. \tag{28}$$

Hence f is increasing throughout the interval. Equations (24) and (26) yield

$$\|B_1 D_0\|_* \le 2d_0 d_1 + z_0 z_1. \tag{29}$$

4.3 Attained global bound

The Bloch block of an arbitrary corrected one-use channel is $T = B_0 + B_1$. Using Eq. (20), operator/nuclear-norm duality, and Eqs. (25) and (29),

$$\begin{aligned}\|T\|_F^2 &= \|CD_0 + B_1\|_F^2 \\ &\le \|D_0\|_F^2 + \|B_1\|_F^2 + 2\|B_1 D_0\|_* \\ &\le 2(d_0 + d_1)^2 + (z_0 + z_1)^2 = 2r + r^2 = \frac{189}{100}.\end{aligned} \tag{30}$$

Theorem 1 (Exact current-bit optimum)**.** *For the ordered instrument in Eqs.* (3)–(4) *at the reference point* (14)*, allowing arbitrary clock-dependent qubit CPTP recoveries conditioned only on the current bit,*

$$\boxed{F_2^M = \frac{289}{400}}. \tag{31}$$

Identity recovery $\mathcal{R}_{k,b} = \mathrm{id}$ *for every* k, b *attains the optimum.*

Proof. Let T_1, T_2 be the two one-clock Bloch blocks. Frobenius Cauchy–Schwarz and Eq. (30) give

$$\mathrm{Tr}(T_2 T_1) = \langle T_2^T, T_1 \rangle_F \le \|T_2\|_F \|T_1\|_F \le \frac{189}{100}. \tag{32}$$

By Eq. (9), every current-bit policy therefore has $F_e \le (1 + 189/100)/4 = 289/400$.

For identity recovery, each one-clock block is the uncorrected GAD block

$$T_1 = T_2 = D_0 + D_1 = \mathrm{diag}(\sqrt{r}, \sqrt{r}, r). \tag{33}$$

It saturates Eq. (32). The identities are exact CPTP maps, with the single Kraus operator I, so the upper bound is attained. □

Combining Theorem 1 with the feasible history value gives

$$H\left(\frac{18}{25}, \frac{3}{10}\right) - F_2^M = \frac{1831}{2500} - \frac{289}{400} = \frac{99}{10000}. \tag{34}$$

This is the exact advantage of the fixed witness over the optimal current-bit controller. Since F_2^B may exceed the witness, Eq. (34) implies $F_2^B - F_2^M \ge 99/10000$ but does not determine that optimum gap. At one use the current-bit and full-history information sets coincide. Thus the earliest depth at which retained coarse history can help is exactly two uses.

5 A uniform rational robustness rectangle

The sharp Gram argument above is specialized to the reference point. A simpler trace-norm bound is weaker there but holds throughout a parameter region. For one corrected use, direct calculation gives

$$\begin{aligned}\mathcal{I}_0(X) &= psX, & \mathcal{I}_1(X) &= qsX, \\ \mathcal{I}_0(Y) &= psY, & \mathcal{I}_1(Y) &= qsY, \\ \mathcal{I}_0(Z) &= \mathrm{diag}(p, \gamma - p), & \mathcal{I}_1(Z) &= \mathrm{diag}(q - \gamma, -q).\end{aligned} \tag{35}$$

Hermitian trace-norm contraction after each branch, followed by the triangle inequality, bounds the columns of every attainable one-clock block T by

$$\|T\boldsymbol{e}_x\|_2, \|T\boldsymbol{e}_y\|_2 \le \sqrt{1 - \gamma}, \qquad \|T\boldsymbol{e}_z\|_2 \le z(p, \gamma), \tag{36}$$

where

$$z(p,\gamma) = \frac{1 + |p-\gamma| + |1-p-\gamma|}{2}. \tag{37}$$

The factorization (6), Eq. (9), and Frobenius Cauchy–Schwarz then give every current-bit policy the global bound

$$F_2^M(p,\gamma) \le U(p,\gamma) := \frac{1 + 2(1-\gamma) + z(p,\gamma)^2}{4}. \tag{38}$$

No unitality, unitarity, or stationarity assumption enters this bound.

Theorem 2 (Uniform history advantage)**.** *For every (p,γ) in the closed rectangle* R *of Eq. (2), the channel in Eq. (3) remains full Choi rank and*

$$F_2^B(p,\gamma) - F_2^M(p,\gamma) \ge \frac{1}{1000}. \tag{39}$$

The same policy (10) is feasible at every point.

Proof. On all of R,

$$1 - p \le \gamma \le p, \tag{40}$$

including equality $1 - p = \gamma$ at the lower-left corner. Thus Eq. (37) reduces, on the entire closed rectangle, to $z = p$. The difference between the feasible history value (13) and the upper bound (38) is consequently

$$G(p,\gamma) := H - U = \frac{1 - 2\gamma + 2p\gamma^2 - p^2}{4}. \tag{41}$$

Its partial derivatives have uniform exact signs:

$$\partial_p G = \frac{\gamma^2 - p}{2} < 0, \qquad p - \gamma^2 \ge \frac{618191}{10^6}, \tag{42}$$

$$\partial_\gamma G = p\gamma - \frac{1}{2} < 0, \qquad \frac{1}{2} - p\gamma \ge \frac{280931}{10^6}. \tag{43}$$

Therefore the minimum lies at the upper-right corner, where exact arithmetic gives

$$\min_{\mathsf{R}} G = G\left(\frac{723}{1000}, \frac{303}{1000}\right) = \frac{2013407}{2000000000} = \frac{1}{1000} + \frac{13407}{2000000000}. \tag{44}$$

Since $F_2^B \ge H$ and $F_2^M \le U$, this proves Eq. (39).

Finally, p, q, γ are strictly positive throughout the rectangle, and

$$pq\gamma^2 \ge \frac{168427911}{10^{10}} > 0. \tag{45}$$

The Kraus-independence argument from Sec. 2 therefore proves uniform full Choi rank. □

The lower-left point of R lies on an absolute-value chamber boundary; the proof includes it algebraically rather than by an open-neighbourhood continuity argument. The rectangle is deliberately conservative. For example, the larger box $[71/100, 29/40] \times [29/100, 61/200]$ stays in the same chamber, but the right-hand side of Eq. (41) is $-591/3200000$ at its upper-right corner. This only shows failure of the sufficient bound there; it is not evidence that the true history advantage is nonpositive.

6 Discussion and limitations

At the reference channel, identity is globally optimal among all current-bit recoveries, even though an environment bit is supplied after each use. Retaining the first bit changes the information structure: the simple bit-flip policy (10) exceeds the exact current-bit optimum. The mechanism is coarse graining. Each disclosed operation contains two hidden Kraus alternatives, so results based on resolved individual outcomes do not collapse this model to the known qubit case [1, 2].

The result is finite dimensional and exact. It uses no perturbative expansion, sampling argument, or numerical optimizer. Nonunital recovery translations are included in the branch matrices B_b and disappear from the final fidelity only through the exact trace-preserving Bloch identity. The rectangle theorem further shows that the separation is not an isolated rational coincidence: the same witness beats every admissible current-bit policy by a fixed amount on a two-parameter closed set.

Several boundaries matter. First, 99/10000 is the exact advantage of the displayed history witness over F_2^M, not the exact value of $F_2^B - F_2^M$. Second, the off-centre current-bit optimum is not known; Theorem 2 uses the upper bound U. Third, neither the rectangle nor the pairing 03|12 is optimized. Other coarse grainings, correlated environments, more than two uses, persistent quantum memory, and experimental implementation are outside the claim. Finally, the literature comparison is bounded rather than an exhaustive priority determination; the author should extend it before submission.

Reproducibility

The accompanying source bundle contains independent direct-Kraus and Liouville reconstructions, an exact checker for the centre certificate, and a separate exact checker for the robustness rectangle. Rational and symbolic arithmetic establish the reported constants. Floating-point calculations are limited to convention and residual checks and are not used to prove either theorem. No empirical data were used. The included reproducibility guide gives the interpreter, dependency versions, and commands.

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